

Quantifying the Cost of Content-Blindness in LLM Inference Scheduling

Kevin Andrew A

Dept. of Information Technology
Loyola-ICAM College of Engineering and Technology
Chennai, India
kevinandrew2559@gmail.com

Kavita Sri

Dept. of Information Technology
Loyola-ICAM College of Engineering and Technology
Chennai, India
kavitasri.27it@licet.ac.in

Abstract—Large language model (LLM) inference services schedule requests without knowing how many tokens each will generate. Because a long response occupies a batch slot for many iterations, first-come-first-serve (FCFS) scheduling suffers head-of-line blocking, and a substantial literature therefore predicts output length from the prompt text in order to approximate shortest-job-first. Reading prompt content is precisely what a privacy-constrained or regulated serving platform cannot do. We quantify what that restriction costs. Analysing 44.1 million requests from Microsoft Azure’s production LLM inference traces, we observe that service work splits into a prefill component, exactly observable at admission from the context length, and a decode component, not observable until the request completes, and that the observable component is the larger one: prefill is 91.2% and 98.7% of attributable marginal work and 76% and 92% of measured engine time at high load. We evaluate CB-SJF-Work, which orders admissions by estimated total service work using only token counts and arrival metadata. Over 20 (conversation) and 11 (code) windows sampled across a full production week, it reduces mean normalised latency by 51.5% and 58.8% relative to FCFS, recovering 90.3% and 79.3% of the improvement attainable by an output-length oracle. Two findings qualify this. The benefit is not free: shortest-first ordering defers long requests, and 99th-percentile latency degrades by $1.9\times$ and $3.1\times$ on average at the highest load, and by up to $4.6\times$ and $6.1\times$ in the worst window, a cost an oracle scheduler also incurs. And a learned content-blind length predictor contributes almost nothing over ordering by raw context length, reaching significance only at the highest load. For these workloads the information a scheduler is denied is largely the information it least needs.

Index Terms—LLM inference serving, cloud scheduling, head-of-line blocking, privacy-preserving systems, production trace analysis, big data architectures

I. INTRODUCTION

Serving large language models is now a dominant consumer of cloud GPU capacity. A modern inference engine such as vLLM [1] or Orca [2] executes requests with *continuous batching*: many requests share a batch, each scheduling iteration emits one token for every resident request, and a request holds its slot and its share of the key-value (KV) cache until it produces its final token.

This creates a scheduling problem with an unusual information structure. How long a request holds its slot is proportional to its output length, which is unknown when the request arrives. Under first-come-first-serve (FCFS), the default in production engines, one long-generating request admitted early

blocks many short ones behind it, producing head-of-line blocking and long tails.

The established response is to predict output length and approximate shortest-job-first (SJF). S^3 fine-tunes a Distil-BERT model on the prompt to predict sequence length [3]; Fu *et al.* learn to rank requests by predicted output length, again from prompt text [4]. Both report large gains, and both require the scheduler to read the user’s prompt. That is a genuine obstacle: a platform under data-protection obligations or offering confidential-computing guarantees may be unable to route prompt content into a scheduling component, and running an auxiliary transformer over every prompt consumes the accelerator capacity the scheduler is trying to conserve. Hence our question: how much of the achievable scheduling benefit remains available to a scheduler that never reads prompt content?

We answer it empirically on Microsoft Azure’s public production LLM inference traces [5], which contain 44.1 million real requests across a conversation workload and a code workload. Our central finding is that the question’s premise, that the valuable information is locked inside the prompt, does not hold for these workloads. Service time has two components: a prefill pass over the input tokens, whose cost is fixed by the context length and therefore visible at admission without reading any content, and a sequence of decode iterations, whose cost is not. On these traces prefill is the larger component under either of the two accountings we report in Section III.

We make the following contributions.

- 1) A quantification of what content-blindness costs an admission scheduler. Ordering by observable work recovers 90.3% (conversation) and 79.3% (code) of the improvement an output-length oracle attains, evaluated over 20 and 11 windows sampled across a full production week (Sections V–VI).
- 2) A characterisation of the observable and unobservable components of service work on 44.1M production requests, reported under two accountings so the conclusion does not rest on one cost convention, together with the finding that output length is weakly predictable from content-blind metadata for conversational traffic

($\rho_s = 0.365$) and effectively unpredictable for code completion ($\rho_s = 0.107$) (Section III).

- 3) Two negative results. A learned content-blind length predictor adds almost nothing over ordering by raw context length, and at high load ordering by predicted output length alone is worse than FCFS on the code workload. Under a content-blindness constraint, the effort of learning to predict output length is not repaid.
- 4) An accounting of the cost side. Reordering admissions degrades tail latency, and controls show this to be the price of departing from arrival order rather than of shortest-first ordering or of content-blindness.

II. BACKGROUND AND RELATED WORK

Continuous-batching inference engines. Orca [2] introduced iteration-level scheduling and vLLM [1] added paged KV-cache management, together establishing the execution model we assume: a batch whose membership changes every iteration, bounded by GPU memory available for the KV cache. Sarathi-Serve [6] chunks prefills to reduce decode stalls, and Splitwise [7] and DistServe [8] disaggregate the two phases onto separate hardware. These change where and how prefill and decode execute; they are orthogonal to which request to admit next, which is our subject.

Output-length prediction for scheduling. S^3 [3] predicts output length with a fine-tuned DistilBERT to reserve KV memory more tightly. Fu *et al.* [4] observe that exact lengths are unnecessary and learn only the relative rank of output lengths, reporting substantial latency reductions. Both extract their signal from prompt text, which is the constraint we remove.

Scheduling on input length. We are not the first to key scheduling on input length. FastServe’s skip-join MLFQ [9] assigns each arrival an initial queue from its profiled prefill time, and absorbs the unknown output length through iteration-level preemption rather than prompt inspection. PecSched [10] analyses an Azure LLM inference trace, reports the same skewed input-length distribution we observe, and preemptively prioritises short-input requests at cluster scale. Both are prior art for content-blind, input-length-keyed scheduling, and both obtain their tail behaviour from preemption, which our non-preemptive setting excludes. Neither quantifies what the unobservable component costs, which is our subject.

Production trace analysis. The Azure LLM inference traces [5] were released alongside Splitwise [7] and DynamoLLM [11]. Splitwise already reports that the coding workload is prompt-dominated, with a median of 13 generated tokens, and we do not claim that observation as new. What we add is its consequence for admission scheduling: how the observable and unobservable components of service work divide, and what a scheduler forfeits by seeing only the former.

III. CHARACTERISING PRODUCTION LLM TRAFFIC

A. Traces

We use the Azure LLM Inference Trace 2024 [5], released under CC-BY-4.0/MIT. It contains two one-week traces from

Microsoft’s production inference service: a conversation workload (27.3M requests) and a code workload (16.8M requests). Each record gives an arrival timestamp, the input context length in tokens, and the number of tokens generated. Crucially for our purpose, the traces contain no prompt text: they are themselves a content-blind view of the workload, which is exactly the information a privacy-constrained scheduler would possess.

B. The observable component is the larger one

In a continuous-batching engine, servicing a request costs a prefill pass proportional to its context length x_i , plus y_i decode iterations where y_i is its output length. Charging decode at its marginal per-token cost (the per-iteration base cost is a property of the batch, not of any one request), the work attributable to request i is

$$W_i = c_f + c_p x_i + c_d y_i, \quad (1)$$

where c_p is the per-token prefill cost, c_d the marginal per-token decode cost, and c_f a fixed overhead. The first two terms are known at admission; only the third requires knowing y_i .

Evaluated over both full traces, (1) puts prefill at 91.2% of attributable marginal work in the conversation trace and 98.7% in the code trace. The reason is the traffic shape. Mean context lengths are 1632 and 2511 tokens against mean output lengths of only 106 and 23, giving aggregate context-to-output token ratios of 15.5:1 and 110.7:1. That ratio is a property of the traffic and involves no cost model at all.

Equation (1) is an attribution convention, not a measurement of engine time. We also report the other natural accounting: the fraction of busy time our simulated engine spends in prefill steps. That fraction is load-dependent, because larger batches amortise the base decode cost over more requests. At $\rho = 0.60$ it is 53.9% (conversation) and 81.8% (code); at $\rho = 0.92$ it is 75.6% and 92.4%. The two accountings differ, most visibly for conversation at low load, and we report both rather than the more favourable one. Under either, prefill is at least half of the work and the component hidden from the scheduler is the smaller one.

C. Output length is not equally predictable

The unobservable component is also unequally guessable. Binning by context-length decile, mean output length rises roughly ninefold across the conversation deciles, while for code the relationship is flat and output length is negligibly correlated with input length. A content-blind predictor (Section IV) attains a rank correlation of $\rho_s = 0.365$ with true output length on conversation but only 0.107 on code, with pairwise ordering accuracies of 0.633 and 0.537 against a chance level of 0.5.

The two workloads therefore have opposite risk profiles: code has both the heavier output-length tail and the weaker predictive signal, so a method depending on accurate length prediction is weakest where it is most needed.

IV. CONTENT-BLIND WORK-AWARE SCHEDULING

A. Threat model and information constraint

The scheduler may observe, per arriving request, the context length in tokens, the arrival timestamp, the workload class, and aggregate statistics over requests that have already completed. It may not observe prompt or response text. This corresponds to a front end that tokenises and meters requests, which it must do for billing regardless, but forwards content only to the model.

B. Policy

Section III implies that ordering by predicted output length optimises the wrong quantity. What determines contention is total service work W_i , and by (1) the observable terms dominate it. CB-SJF-Work therefore admits requests in increasing order of

$$\widehat{W}_i = c_f + c_p x_i + c_d \hat{y}_i, \quad (2)$$

where $\hat{y}_i$ is a content-blind estimate of output length. Because $c_d \hat{y}_i$ contributes only a small share of $\widehat{W}_i$, the policy degrades gracefully: when $\hat{y}_i$ is uninformative, the ranking is still governed by the exactly-known prefill term.

CB-SJF-Work is not offered as a novel mechanism. It is the natural content-blind policy, used here as an instrument: placed between FCFS and an oracle, it measures what the information constraint costs.

We obtain $\hat{y}_i$ from a gradient-boosted regression tree trained on $\log(1 + y)$ over ten content-blind features: context length and its logarithm, workload class, hour of day, day of week, inter-arrival time, recent arrival rate, and rolling mean and 90th-percentile output length and mean context length over the previous 512 completed requests. Rolling statistics are shifted by one request, so no request observes its own or any future outcome, and an automated check fails the run if any feature exceeds 0.95 rank correlation with the target. The predictor trains on a contiguous earlier period of the same trace, disjoint from every evaluation window, and is applied unchanged thereafter.

Admission is head-of-line: if the highest-priority waiting request does not fit in the KV cache, the scheduler waits rather than scanning past it. The KV budget for a model with L layers, H KV heads and head dimension d at b bytes per element is

$$B = \lfloor (M_{\text{gpu}} - M_w) / (2LHdb) \rfloor \quad (3)$$

tokens, where M_w is the weight footprint. Note that the KV reservation in Algorithm 1 charges the true output length y_i , not the estimate $\hat{y}_i$. This is a simplification: reserving on an underestimate would require the preemption and recompute path that real engines use when a reservation is exceeded, which we do not model. It applies identically under every policy, including FCFS and both oracles, so it confers no advantage on any of them. The scheduler’s ordering key, which is what this paper studies, remains strictly content-blind.

Algorithm 1 CB-SJF-Work admission (one scheduling iteration)

Require: pending min-heap Q , running set R , KV budget B , used u

- 1: **on arrival of** i : push i into Q with key $\widehat{W}_i$ from (2) $\triangleright$ content-blind
- 2: **while** $Q \neq \emptyset$ **do**
- 3: $i \leftarrow \text{PEEK}(Q)$
- 4: **if** $|R| < B_{\text{max}}$ **and** $u + x_i + y_i \leq B$ **then**
- 5: $\text{POP}(Q)$; $R \leftarrow R \cup \{i\}$; $u \leftarrow u + x_i + y_i$
- 6: **else**
- 7: **break** $\triangleright$ head-of-line; do not scan past
- 8: **end if**
- 9: **end while**
- 10: run prefill for newly admitted requests, else one decode iteration
- 11: evict completed requests from R and release their KV

V. EXPERIMENTAL METHODOLOGY

Simulator. We simulate one continuous-batching replica with iteration-level scheduling and the KV bound of (3). Prefill cost is affine in prompt tokens, decode cost affine in batch size, with parameters for a 13B-parameter model ($L = 40$, $H = 40$, $d = 128$, fp16) on an 80 GB accelerator, giving $B = 65,917$ tokens. The cost model is analytical, not measured on our hardware; Section VII sweeps it.

Load calibration. The published traces are fleet-aggregate (75–87 req/s), far above single-replica capacity. We therefore measure each window’s saturated throughput μ by driving the engine with an unbounded backlog, then rescale the arrival timeline so that $\rho = \lambda/\mu$ takes the values $\{0.60, 0.75, 0.85, 0.92, 0.98\}$. This preserves the arrival process shape and request mix up to an affine rescaling of the clock. Measured capacity is 1.79 req/s (conversation) and 1.41 req/s (code).

Two caveats apply, because ρ is easy to misread. First, μ is measured at maximum concurrency where the per-iteration base cost amortises over a full batch, whereas the runs operate at smaller batches, so measured utilisation exceeds ρ : 0.91 (conversation) and 0.71 (code) at $\rho = 0.60$. Second, this does not bias the comparison: throughput agrees across all eight policies to within 0.07% at every load.

Windows. We evaluate the first 20,000 requests of each of 20 (conversation) and 11 (code) windows sampled at even intervals across the full week, each at least two hours from the next and after the predictor’s training period. Twenty thousand requests span 3.7 to 19.9 minutes of arrivals, so a window is a few minutes of traffic rather than a 20-minute period. We apply no busyness ranking, because rescaling to a target ρ normalises a window’s arrival rate away and window choice affects only the request mix. We do require each window to hold at least 20,000 requests so every run has the same length, which is implicitly a traffic threshold: it excludes no conversation candidates but 10 of 24 code candidates, the

quietest periods of that trace. Replaying the same windows under every policy makes all comparisons paired; the leading 10% of each run is warm-up.

Policies. We compare FCFS (the engine default); SJF-Ctx, ordering by context length alone; SJF-Pred, ordering by predicted output length; our CB-SJF-Work; two controls, LJF-Work (the CB-SJF-Work key negated) and Random, which isolate ordering direction from the information used; and two oracles, Oracle-SJF and Oracle-Work, ordering by true output length and true total work. The oracles need information unavailable at admission and are upper bounds, not deployable systems.

Metrics. Mean normalised latency $\bar{L}_n$ (end-to-end latency per output token), 99th-percentile end-to-end latency, and service-level objective (SLO) attainment, the fraction of requests meeting both a 2 s time-to-first-token (TTFT) and a 100 ms time-between-tokens (TBT) target.

VI. RESULTS

A. Content-blind scheduling captures most of the benefit

Table I reports the contended operating point $\rho = 0.92$. CB-SJF-Work reduces mean normalised latency from 1161 to 506 (conversation) and from 16724 to 6327 (code). Fig. 1 shows the full load sweep. Below $\rho \approx 0.75$ all policies nearly coincide, because the admission queue is shallow and there is little ordering decision to make; we report this rather than restricting the evaluation to loads that flatter the method. Above that point the policies separate.

Table II quantifies how much of the attainable benefit survives the content-blind restriction. Averaged over $\rho \geq 0.85$, CB-SJF-Work recovers 90.3% of the Oracle-Work improvement on the conversation workload and 79.3% on code. The improvement is positive in every window at every load above $\rho = 0.75$ (one-sided Wilcoxon signed-rank, $p < 10^{-3}$).

Two cautions bear on reading these percentages. The headline is a ratio of means, so it is weighted by whichever window has the worst FCFS run; the per-window range in Table II is wide (6% to 77% on conversation at $\rho \geq 0.85$) and the mean of per-window improvements is 47.0%, not 51.5%. And the recovered-gap percentage is a ratio of two quantities that both approach zero at low load, so its high value at $\rho = 0.60$ describes a setting where the absolute improvement is under 1%.

A third concern deserves a direct test. Normalised latency divides by output tokens, so for a one-token request it degenerates to raw latency, and short requests could dominate the mean in a way that rewards shortest-first ordering by construction. They do carry it disproportionately: at $\rho = 0.92$ requests emitting at most four tokens are 10.9% and 36.5% of arrivals but 50.2% and 78.6% of total normalised latency. Restricted to requests emitting at least eight tokens, the improvement at $\rho \geq 0.85$ is 47.04% and 48.5% against 51.5% and 58.8%, positive in all 20 and 11 windows, though the inflation grows as load falls: at $\rho = 0.75$ the restricted improvement on code is 11.1% against 22.9%. The headline is inflated by short requests but not an artefact of them.

B. The mean-latency gain is paid for in the tail

Shortest-first ordering reduces the mean by deferring long requests, so it must lengthen the tail. We report that cost rather than leave it to be inferred from Table I. On code, CB-SJF-Work has a worse 99th-percentile latency than FCFS in 31 of 33 window-load cells at $\rho \geq 0.85$, by a factor of $2.05\times$ averaged over those three loads and by up to $6.06\times$ in the worst single window. On conversation the tail is essentially unchanged up to $\rho = 0.92$ (mean $0.98\times$ and $1.00\times$, though still worse than FCFS in 12 of 20 windows at $\rho = 0.92$) and degrades to a mean of $1.86\times$ at $\rho = 0.98$, worse in all 20 windows, with a worst case of $4.57\times$. Means understate this cost considerably and we give both.

This cost is not attributable to content-blindness. All ratios below, and in Table II, are means of per-window ratios. An oracle with perfect output-length knowledge still degrades the tail, by $1.31\times$, $1.65\times$ and $2.78\times$ on code at the three highest loads: marginally less than our $1.33\times$, $1.70\times$ and $3.12\times$, but nowhere near eliminating the penalty. Oracle-SJF, ordering purely by true output length, is far worse (1261 s against our 556 s at $\rho = 0.92$).

Nor is it a property of shortest-first ordering, as we first assumed. Two controls separate ordering direction from information: LJF-Work negates the CB-SJF-Work key, so it is content-blind in the same way but longest-first, and Random orders by an uninformative key. At $\rho = 0.92$ their P_{99} ratios against FCFS are $7.51\times$ and $3.25\times$ (conversation) and $8.82\times$ and $4.23\times$ (code), against $1.00\times$ and $1.70\times$ for CB-SJF-Work. Random reordering, which is not shortest-first at all, damages the tail more than we do; since FCFS and Random differ only in whether the key is monotone in arrival time, this isolates starvation as the mechanism. The penalty is therefore the price of departing from arrival order at all, and of the three reorderings we test, shortest-first is the least harmful. LJF-Work is also $5.3\times$ worse on the mean (2688 against 506), though its tail magnitude is amplified by head-of-line admission, which fit-tests the largest pending request first. An operator who cares primarily about the tail should not reorder without an anti-starvation mechanism [9], [10].

C. The learned predictor earns almost nothing

CB-SJF-Work differs from SJF-Ctx only by the $c_d \hat{y}_i$ term. Its contribution ranges from -0.11% to $+0.39\%$ on conversation and -0.06% to $+3.45\%$ on code, is negative in 5 of the 10 cells, and reaches $p < 0.05$ only at the highest load on either workload (code $p = 0.009$, 9 of 11 windows; conversation $p = 0.027$, 13 of 20). At $\rho = 0.92$ the two differ by 506 against 505 and 6327 against 6337. Ordering by raw context length is, for practical purposes, as good as ordering by a learned estimate of total work, and we would not recommend deploying the predictor for this purpose.

Ordering by predicted output length alone is worse on code, and we state the strength of that evidence precisely because it favours us. SJF-Pred attains $\bar{L}_n = 23081 \pm 28340$ against 16724 for FCFS at $\rho = 0.92$, with P_{99} of 1882 s against 271 s, exceeding FCFS in 8 of 11 windows (one-sided Wilcoxon $p =$

TABLE I

SCHEDULING PERFORMANCE AT OFFERED LOAD $\rho = 0.92$, AVERAGED OVER 20 (CONVERSATION) AND 11 (CODE) DISJOINT WINDOWS SAMPLED ACROSS THE FULL WEEK (MEAN $\pm$ S.D.). $\bar{L}_n$ IS MEAN NORMALISED LATENCY (MS PER OUTPUT TOKEN); P_{99} IS 99TH-PERCENTILE END-TO-END LATENCY (S); SLO IS THE FRACTION OF REQUESTS MEETING BOTH THE TTFT AND TBT TARGETS. NOTE THE TAIL COST: ON CODE EVERY REORDERING DEGRADES P_{99} RELATIVE TO FCFS, WHILE ON CONVERSATION AT THIS LOAD IT IS ESSENTIALLY UNCHANGED. SECTION VI ISOLATES THE CAUSE. [†]ORACLE POLICIES REQUIRE KNOWLEDGE UNAVAILABLE AT ADMISSION TIME AND ARE UPPER BOUNDS, NOT DEPLOYABLE SYSTEMS.

Policy	Conversation			Code		
	$\bar{L}_n$	P_{99} (s)	SLO (%)	$\bar{L}_n$	P_{99} (s)	SLO (%)
FCFS (engine default)	1161 $\pm$ 921	104	18.8	16724 $\pm$ 16657	271	8.6
SJF-Ctx	505 $\pm$ 254	103	34.5	6337 $\pm$ 5519	562	15.4
SJF-Pred	529 $\pm$ 240	207	33.0	23081 $\pm$ 28340	1882	12.2
CB-SJF-Work (ours)	506 $\pm$ 253	104	34.5	6327 $\pm$ 5542	556	15.4
Oracle-SJF [†]	287 $\pm$ 23	129	31.9	2289 $\pm$ 669	1261	14.4
Oracle-Work [†]	436 $\pm$ 184	102	34.4	3676 $\pm$ 2769	528	15.3

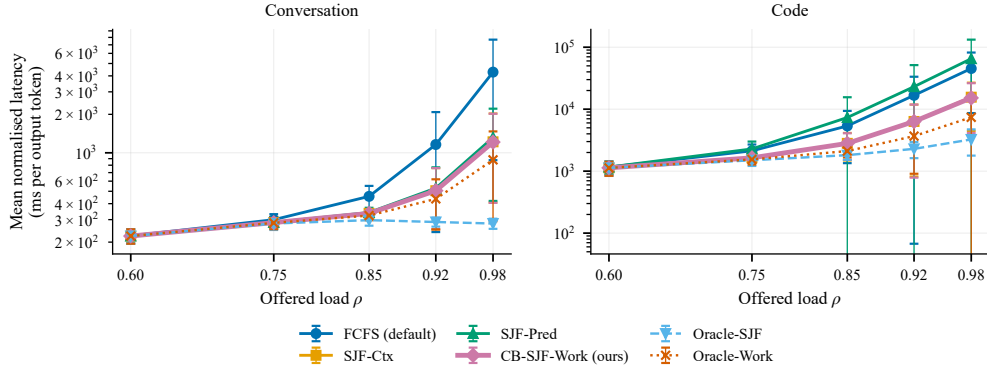


Fig. 1. Mean normalised latency against offered load, over 20 (conversation) and 11 (code) windows sampled across a full production week (error bars: s.d.). Below $\rho \approx 0.75$ the queue is shallow and all policies nearly coincide. CB-SJF-Work tracks Oracle-Work closely and is visually indistinguishable from SJF-Ctx; SJF-Pred, which orders by predicted output length, diverges and at high load on the code workload is worse than FCFS.

TABLE II

CB-SJF-WORK RELATIVE TO FCFS AGAINST OFFERED LOAD. ROWS PER WORKLOAD: MEAN-LATENCY IMPROVEMENT (%); ITS PER-WINDOW RANGE (%); PERCENTAGE OF THE ORACLE-ATTAINABLE IMPROVEMENT RECOVERED; AND THE RATIO OF 99TH-PERCENTILE LATENCY TO FCFS, WHERE A VALUE ABOVE 1.0 MEANS THE TAIL IS WORSE. THE FIRST THREE ROWS ARE THE BENEFIT, THE FOURTH IS THE COST.

Workload	Quantity	Offered load ρ				
		0.60	0.75	0.85	0.92	0.98
Conv.	improvement	0.4	5.1	26.4	56.5	71.7
	range	−0–2	1–14	6–44	16–69	39–77
	oracle gap	96.3	92.4	90.4	90.4	90.2
	P_{99}/FCFS	1.00	1.00	0.98	1.00	1.86
Code.	improvement	2.7	22.9	47.7	62.2	66.5
	range	1–12	4–43	14–61	27–71	53–73
	oracle gap	93.8	83.8	78.6	79.7	79.4
	P_{99}/FCFS	1.01	1.05	1.33	1.70	3.12

0.06) and 9 of 11 at $\rho = 0.98$ ($p = 0.03$). The direction is consistent but the magnitude is not tightly estimated. On conversation, where the predictor has some signal, SJF-Pred beats FCFS on mean latency in all 20 windows yet doubles P_{99} (207 s against 103–104 s). Optimising a quantity that cannot be estimated adds ordering noise, and a static priority from a near-random key starves whatever it ranks last. This result

is scoped to the content-blind setting: it does not evaluate the prompt-reading predictors of S^3 [3] or Fu *et al.* [4], which use a signal ours cannot see.

D. SLO attainment

CB-SJF-Work roughly doubles SLO attainment over FCFS, 34.5% against 18.8% (conversation) and 15.4% against 8.6% (code), matching Oracle-Work to within 0.1 points; SJF-Ctx matches both, so here too the gain comes from ordering by context length. Attainment falls steeply with load, from 54.3% to 31.1% and 34.5% to 13.4%, so contention accounts for much of it. It is not the whole story: even at our lowest load 46% and 66% of requests miss the target, because one unchunked prefill step for a multi-thousand-token prompt consumes much of the 2 s time-to-first-token budget before any queueing. The policy comparison stands; the absolute level is not a claim about production.

VII. LIMITATIONS AND THREATS TO VALIDITY

The two sweeps below use the first eight evaluation windows per workload rather than all 20 and 11. Their baseline therefore differs from Table I: the ratio-of-means estimator is sensitive to window membership, giving 46.0% at $\rho = 0.92$ here against 56.5% there.

The cost model is analytical. We do not have access to the accelerator hardware these traces were served on, and our coefficients are modelled rather than measured. The quantity our mechanism depends on, however, is the ratio of context to output tokens, which is a property of the traces: 15.5:1 and 110.7:1 in aggregate. We therefore stress that ratio directly, by scaling output length, rather than perturbing a cost coefficient. Scaling output length moves both the decode share of work and the KV footprint, which is the binding constraint in our engine; scaling a decode cost coefficient alone leaves the KV footprint untouched and is uninformative.

Longer generations, including reasoning and agentic workloads. Scaling output length up to $16\times$ drives the prefill share of engine time from 75.4% to 11.2% (conversation) and 92.9% to 48.8% (code). The improvement over FCFS does not collapse, but the workloads move in opposite directions: it rises from 46.0% to 75.1% on conversation and declines from 63.3% to 50.5% on code. In neither case does ordering by observable work stop being a good ordering, even where that work becomes a minority of engine time. What degrades in both is the price of content-blindness: CB-SJF-Work runs $1.08\times$ the mean normalised latency of Oracle-Work at $1\times$ but $3.76\times$ and $5.53\times$ at $16\times$, and the learned predictor, worth 0.3% today, becomes worth 21.1%. Our conclusions would need revisiting for long-generation workloads.

Prefix caching. This is the most serious threat and the traces cannot answer it: production stacks serve repeated prefixes from cache, multi-turn conversation has the highest hit rates, and the traces record context length but not cache residency. If we reduce executed prefill to model a hit rate, while the scheduler still observes the full context length (cache-hit length depends on token identity and so is not content-blind), the improvement over FCFS falls to 12.8% (conversation) and 14.8% (code) at a 90% hit rate; on code the decline is a cliff rather than a slope, 48.6% at a 75% hit rate. Measured against an oracle that is itself denied cache residency, the recovered gap appears to hold up, at 85.5% and 74.5%, but that comparison flatters us: both policies then order by the same inflated prefill estimate and converge for that reason. Against a cache-aware oracle, which sees the executed prefill, the recovered gap falls to 66.0% and 69.8%. Prefix-cache-aware content-blind scheduling is the main open problem this work exposes.

Non-preemptive admission. Our scheduler controls admission order only: no preemption, swapping, or aging. This isolates the effect of admission information, and it is also why our tails degrade; preemptive schedulers [9], [10] address that failure mode. Our oracles are upper bounds on admission-order policies. We likewise model a single replica without chunked prefill [6] or phase disaggregation [7], [8]. Chunking does not change total prefill work or KV occupancy, so we expect the central claim to survive, but we have not verified it. Finally, two workloads from one provider over one week bound a claim about the regime we measured, not a universal one. That regime has a context-to-output ratio of 15.5:1 and 110.7:1, and the $16\times$ sweep above is evidence along that axis:

the advantage rises on conversation and falls on code.

Reproducibility. All experiments use public data. Windows are selected deterministically, the pipeline is content-hashed to verify byte-identical reruns, seeds live in one configuration module, and every number in this paper is generated programmatically from stored result files rather than transcribed. Code will be released upon acceptance.

VIII. CONCLUSION

We asked what LLM inference scheduling loses if the scheduler may not read prompt content. On 44.1M requests of Azure production traffic the answer is: less than the literature’s focus on prompt-based length prediction would suggest. Prefill, the component observable without reading content, is the larger part of service work on these traces, and ordering by estimated total work recovers 79.3% to 90.3% of the oracle-attainable improvement in mean normalised latency and roughly doubles SLO attainment, even on a workload where output length is negligibly correlated with anything the scheduler can see.

The practical implication is narrow but useful: on workloads resembling these, a scheduler that never sees prompt content gives up little, and ordering by context length alone captures nearly all of it. The natural next steps are to combine content-blind ordering with preemption or aging, which our tail results show is needed, and to extend it to prefix-cache-aware serving.

REFERENCES

- [1] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, “Efficient memory management for large language model serving with PagedAttention,” in *Proc. 29th ACM Symposium on Operating Systems Principles (SOSP)*, 2023.
- [2] G.-I. Yu, J. S. Jeong, G.-W. Kim, S. Kim, and B.-G. Chun, “Orca: A distributed serving system for transformer-based generative models,” in *Proc. 16th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2022.
- [3] Y. Jin, C.-F. Wu, D. Brooks, and G.-Y. Wei, “ S^3 : Increasing GPU utilization during generative inference for higher throughput,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [4] Y. Fu, S. Zhu, R. Su, A. Qiao, I. Stoica, and H. Zhang, “Efficient LLM scheduling by learning to rank,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [5] Microsoft Azure, “AzurePublicDataset: Azure LLM inference trace 2024,” <https://github.com/Azure/AzurePublicDataset>, 2024, accessed 2026-08-02.
- [6] A. Agrawal, N. Kedia, A. Panwar, J. Mohan, N. Kwatra, B. S. Gulavani, A. Tumanov, and R. Ramjee, “Taming throughput-latency tradeoff in LLM inference with Sarathi-Serve,” in *Proc. 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024.
- [7] P. Patel, E. Choukse, C. Zhang, A. Shah, Í. Goiri, S. Maleki, and R. Bianchini, “Splitwise: Efficient generative LLM inference using phase splitting,” in *Proc. 51st Annual International Symposium on Computer Architecture (ISCA)*, 2024.
- [8] Y. Zhong, S. Liu, J. Chen, J. Hu, Y. Zhu, X. Liu, X. Jin, and H. Zhang, “DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving,” in *Proc. 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024.
- [9] B. Wu, Y. Zhong, Z. Zhang, S. Liu, F. Liu, Y. Sun, G. Huang, X. Liu, and X. Jin, “FastServe: Iteration-level preemptive scheduling for large language model inference,” in *Proc. 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2026.
- [10] Z. Zhang and H. Shen, “PecSched: Preemptive and efficient cluster scheduling for LLM inference,” arXiv:2409.15104, 2024.
- [11] J. Stojkovic, C. Zhang, Í. Goiri, J. Torrellas, and E. Choukse, “DynamoLLM: Designing LLM inference clusters for performance and energy efficiency,” in *Proc. IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 2025.